L9: Summary of SI, SIS, SIR Models with Arbitrary Degree Distribution

Model	Continuum Equation	τ	λ_c
SI	$\frac{di_k}{dt} = k\,\beta\,(1 - i_k)\,\theta_k$	$\frac{\bar{k}}{\beta(\bar{k^2} - \bar{k})}$	-
SIS	$\frac{di_k}{dt} = k\,\beta\,(1 - i_k)\,\theta_k - \mu\,i_k$	$\frac{\bar{k}}{\beta\,\bar{k^2} - (\beta+\mu)\bar{k}}$	$\frac{\bar{k}}{\bar{k^2}}$
SIR	$\frac{di_k}{dt} = k\,\beta\,(1 - i_k - r_k)\,\theta_k - \mu\,i_k$	$\frac{\bar{k}}{\beta\,\bar{k^2} - (\beta+\mu)\bar{k}}$	$\frac{\bar{k}}{\bar{k^2} - \bar{k}}$

Even though we showed the derivations for the density function $\theta(t)$ only for the SIS model, it is simple to write down the corresponding equations for the SI and SIR models.

The table summarizes the differential equation and key results for each of the three models.

The parameter τ is the characteristic timescale.

The SI model always leads to an epidemic. For the two other models, however, the epidemic threshold depends on the ratio of the first two moments of the degree distribution. λ_c is the minimum value of λ for the emergence of an endemic state (only for SIS and SIR).

We suggest that you contrast these results with the corresponding formulas for the case of homogenous mixing.

Food For Thought

Repeat the derivations we performed in this lesson for the SIS model in the case of the SI and SIR models.